

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO5 ASSESSMENT TOOL 1 – RESEARCH PAPER REVIEW / LITERATURE SURVEY*Literature Survey Report*

Course Code	CSA1013	Course Title	Software Engineering
CO Assessed	CO5	Assessment Tool	Tool 1 – Literature Survey
Weightage	20%	Total Marks	25
Problem No.	13		
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CO5 Statement

Intelligent E-Governance Citizen Service Request Management System.

Assigned Problem Statement**Intelligent E-Governance Citizen Service Request Management System**

This report surveys the existing body of research on AI- and computer-vision-based road surface damage detection, drone/UAV-based inspection, and GIS-integrated pavement maintenance planning, in order to understand current techniques, benchmark datasets, performance levels, limitations, and unresolved research gaps relevant to the assigned system.

Code of Conduct

I S SUVIKUMAR / 192424262 (Name / Reg. No.) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

1. Introduction

Road infrastructure deteriorates continuously under traffic loading and environmental exposure, producing cracks, potholes, rutting, and other surface defects that threaten safety and raise vehicle-operating costs. Traditional inspection relies on manual visual surveys, which are slow, subjective, and cannot scale to large road networks. Over the past decade, researchers have applied computer vision, deep learning, drone/UAV imaging, and Geographic Information Systems (GIS) to automate detection, classification, severity estimation, and maintenance planning. This literature survey reviews ten representative, recent studies spanning these four themes — image-based deep-learning detection, UAV-based inspection, GIS/spatial-AI integration, and pavement-condition prediction — to establish the state of the art and identify the research gap that motivates the assigned AI-Based Road Surface Damage Detection and Maintenance Planning System.

2. Paper Selection Methodology

Papers were selected from reputed scholarly sources — including Computer-Aided Civil and Infrastructure Engineering (Wiley), Journal of Real-Time Image Processing (Springer), Scientific Reports (Nature), PLOS ONE, IEEE Xplore, ACM Digital Library, ASCE Journal of Surveying Engineering, and arXiv preprints — using search terms such as "road damage detection deep learning", "pothole detection UAV YOLO", "road crack CNN dataset", and "GIS pavement condition machine learning". Priority was given to papers published between 2018 and 2025 that (a) directly address automated road-defect detection, classification, or severity/condition assessment, and (b) are frequently cited or represent a widely used benchmark dataset or technique in this domain. Ten papers were selected to give balanced coverage across the four themes above.

3. Individual Paper Reviews

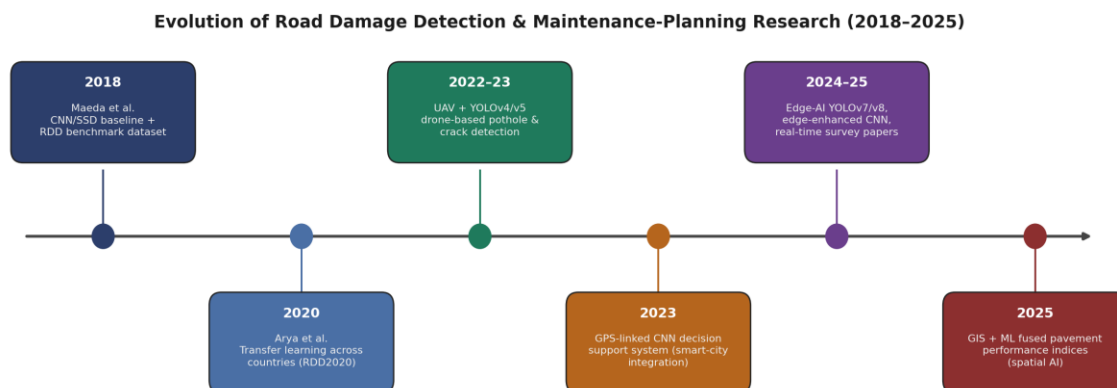


Figure 1: Evolution of road-damage detection and maintenance-planning research relevant to the assigned problem statement

3.1 Maeda et al. (2018) — Road Damage Detection and Classification Using Deep Neural Networks with Smartphone Images

Objective: Create a large-scale, publicly available road-damage image dataset and evaluate deep object-detection models capable of both detecting and classifying damage type from smartphone images.

Methodology: Two single-shot object detectors, SSD Inception V2 and SSD MobileNet, were trained to detect and classify eight damage categories (cracks, potholes, rutting/bumps, etc.) using bounding-box annotations.

Dataset / Tools: A new benchmark of 163,664 road images (9,053 containing cracks, 15,435 annotated damage instances) collected via smartphones mounted on cars across seven Japanese municipalities; this became the widely used RDD/RDD2018 dataset.

Key Findings: SSD MobileNet gave better overall recall than SSD Inception V2 while being roughly twice as fast, and could run on-device on a smartphone with about 1.5-second inference, with precision and recall reported above 75% overall (varying by damage class).

3.2 Arya et al. (2020) — Transfer Learning-Based Road Damage Detection for Multiple Countries

Objective: Extend the original RDD dataset and detection approach to multiple countries and assess how well a single trained model generalizes across different road-surface types and imaging conditions.

Methodology: Object-detection CNNs pretrained on the original Japan dataset were fine-tuned via transfer learning on newly collected data from India and the Czech Republic, producing the multi-country RDD2020 dataset and benchmark models.

Dataset / Tools: RDD2020 — an expanded, multi-country annotated road-damage image dataset combining Japan, India, and Czech Republic data, later used in the IEEE Global Road Damage Detection Challenge (GRDDC).

Key Findings: Transfer learning improved cross-country generalization compared to training on a single country's data alone, but detection accuracy still dropped noticeably when a model trained on one country's road surfaces and imaging setup was applied directly to another.

3.3 Advancements in Real-Time Road Damage Detection — Survey (2025, Journal of Real-Time Image Processing)

Objective: Systematically review deep-learning models and publicly available Road Damage Detection (RDD) datasets, comparing them for real-time deployment suitability.

Methodology: A structured literature survey comparing CNN-based object detectors — YOLO variants, R-CNN family, and SSD — on accuracy, model complexity, and inference speed, alongside an overview of attention mechanisms.

Dataset / Tools: Synthesizes results across the RDD2018, RDD2020, RDD2022, UAV-PDD2023, and other public datasets (Kaggle, Stellenbosch, EGY_PDD, Slovakia potholes).

Key Findings: YOLO-family single-stage detectors generally offer the best accuracy/speed trade-off for real-time road-damage detection, but no single dataset or model yet generalizes well across all road types, weather, and camera setups; dataset privacy handling (face/license-plate blurring) is inconsistent across datasets.

3.4 A Review of Vision-Based Pothole Detection Methods Using Computer Vision and Machine Learning (2024, PMC)

Objective: Review vision-based pothole-detection methods across image-based, 3D point-cloud, and smartphone-sensor approaches, and compare classical machine learning against deep learning.

Methodology: Literature review comparing classical machine-learning classifiers (e.g., SVM) with lightweight CNN architectures (VGG16, ResNet-34, ResNet-50) used to classify distress types such as potholes, cracks, patches, and bleeding.

Dataset / Tools: Draws on multiple public pothole/pavement-distress image datasets used across the reviewed studies rather than a single proprietary dataset.

Key Findings: Lightweight CNN-based multi-label classification (MLC) models achieved up to about 97% accuracy and a 93% F1-score across distress types, showing deep learning's clear advantage over manual or purely geometric 3D-reconstruction methods, though at higher computational cost.

3.5 Real-Time Road Defect Monitoring from UAV Visual Data Sources (2023, ACM)

Objective: Develop and evaluate a UAV (drone)-based object-detection framework for automated road crack and pothole detection and localization.

Methodology: A YOLOv5 object detector was trained on annotated drone-captured images to classify and localize two defect classes — cracks and potholes.

Dataset / Tools: A custom UAV-imagery dataset annotated for two damage classes, collected specifically for this study.

Key Findings: The YOLOv5-based UAV framework achieved an IoU of up to 95.64% for detection and an F1-score of up to 67.82% for classification, demonstrating that drone imagery can extend road-defect detection beyond road-level (vehicle-mounted) coverage to inaccessible or hazardous areas.

3.6 Drone-Based Potholes Detection Using Machine Learning on Various Edge AI Devices in Real-Time (2024, IEEE)

Objective: Compare successive YOLO model versions for real-time pothole detection when deployed directly on resource-constrained Edge AI hardware.

Methodology: YOLOv5, YOLOv7, and YOLOv8 were trained and benchmarked for accuracy, precision, and recall, then deployed on Edge AI devices for on-board, real-time inference from drone video.

Dataset / Tools: A drone-collected pothole-image dataset used consistently across all three YOLO versions for a fair comparison.

Key Findings: YOLOv8 achieved the highest accuracy (78%), followed by YOLOv5 (70%) and YOLOv7 (67%), showing that newer YOLO architectures are not always strictly better for a given edge-hardware/dataset combination and that model choice should be validated per deployment scenario.

3.7 Road Pothole Detection Using Unmanned Aerial Vehicle Imagery and Deep Learning Technique (2022, Zanco Journal)

Objective: Evaluate whether UAV-derived orthophotos can support accurate, efficient pothole detection compared to ground-based visual inspection.

Methodology: YOLOv4 and its lightweight variant YOLOv4-tiny were trained on UAV-derived road orthophotos (stitched aerial imagery) and compared on detection accuracy and training/inference speed.

Dataset / Tools: 5,300 UAV-derived pothole images (90% train / 10% test) from two selected road study areas.

Key Findings: YOLOv4 reached a higher mAP (91.2%) than YOLOv4-tiny (85.7%), but YOLOv4-tiny trained roughly $3.6\times$ faster per iteration and ran inference faster, illustrating the classic accuracy-versus-efficiency trade-off for UAV-based detection.

3.8 Chu et al. (2023) — *Deep Learning Method to Detect the Road Cracks and Potholes for Smart Cities*

Objective: Design an end-to-end decision-support system (DSS) that not only detects road cracks/potholes but also geo-tags and routes them to a road-management information system for maintenance action.

Methodology: A layered architecture: an image-capture layer feeds a CNN-based Pothole/Crack Detection (PCD) decision layer, whose output is combined with GPS coordinates and forwarded to a road-management information system that informs both drivers and maintenance departments.

Dataset / Tools: Road-surface images paired with GPS coordinates, used to train and evaluate the CNN-based PCD decision layer within the proposed DSS.

Key Findings: Coupling CNN-based detection directly with GPS tagging and a maintenance-information layer demonstrated the practical feasibility of an integrated detect-to-notify pipeline, closely mirroring the objectives of the assigned system, though the study's GIS component remained relatively basic (coordinate tagging rather than full spatial analytics).

3.9 Al-Kindi et al. (2025) — *Utilizing Spatial Artificial Intelligence to Develop Pavement Performance Indices (Scientific Reports / Nature)*

Objective: Merge machine-learning classification with Geographic Information Systems (GIS) to evaluate and visualize road condition using the Pavement Condition Index (PCI) and International Roughness Index (IRI).

Methodology: ML classification models were integrated with a GIS platform so that predicted pavement-performance indices could be spatially mapped and analyzed across a road network, rather than reported only as tabular scores.

Dataset / Tools: Road-segment-level condition survey data (PCI/IRI measurements) linked to GIS spatial road-network data for the case-study region.

Key Findings: The fused ML+GIS approach produced spatially interpretable pavement-performance maps that conventional, non-spatial ML models could not, supporting more targeted maintenance planning — directly validating the value of GIS integration for the assigned system's mapping and prioritization objectives.

3.10 Wang et al. (2025) — *Optimizing CNN for Pavement Distress Detection via Edge-Enhanced Multi-Scale Feature Fusion (PLOS ONE)*

Objective: Improve CNN-based crack-classification accuracy under heterogeneous, noisy real-world pavement backgrounds by better preserving crack-edge detail.

Methodology: An Edge-Enhanced Multi-Scale Feature Fusion (EE-MSFF) mechanism was introduced into a deep CNN architecture, combining multi-scale feature maps with explicit edge-detection cues (building on prior work combining CNNs with Laplacian-of-Gaussian and SUSAN-filter edge detection).

Dataset / Tools: Pavement-surface distress images with heterogeneous, complex backgrounds representative of real-world road environments.

Key Findings: The edge-enhanced multi-scale fusion approach improved discrimination between true cracks and background noise/non-crack samples compared to standard CNN baselines, indicating that architectural refinements — not just larger datasets — can meaningfully improve detection reliability.

4. Comparative Analysis

The reviewed studies differ substantially in detection approach (single-stage YOLO/SSD vs. two-stage R-CNN family vs. classical ML), imaging platform (smartphone/vehicle-mounted vs. UAV/drone vs. static survey imagery), and reported metric (accuracy, mAP, F1-score, IoU), which makes direct numeric comparison imperfect. The chart below places the reported top-line performance figures side by side for reference, while the comparative table further below adds methodology, dataset, limitations, and gap for each paper.

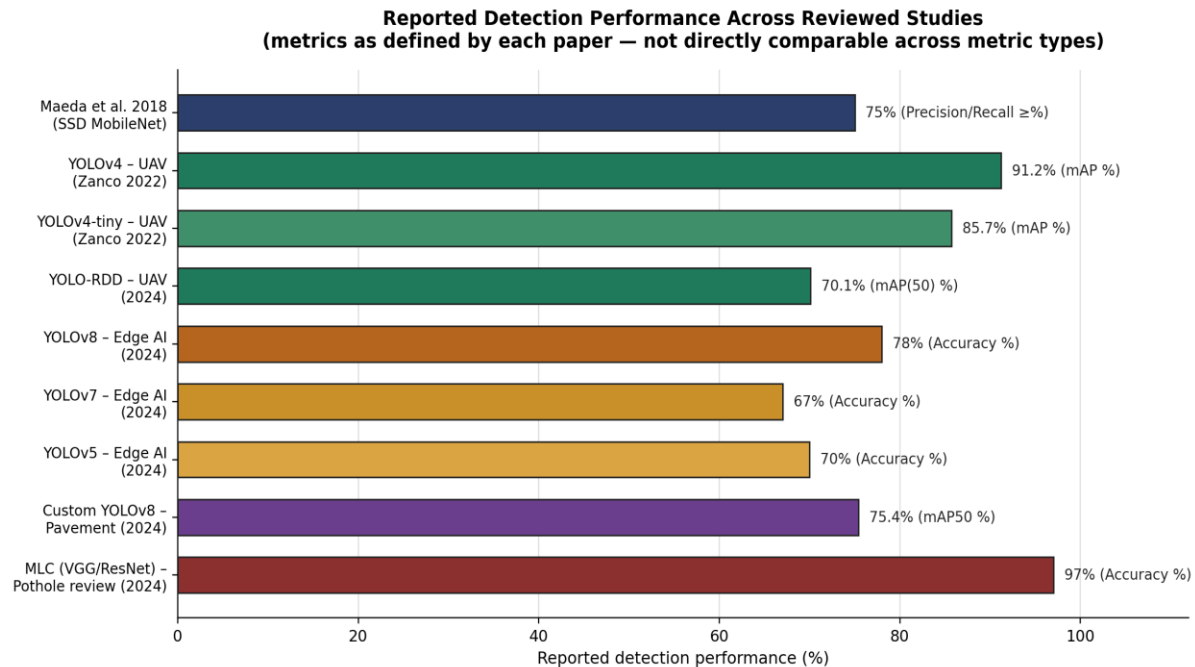


Figure 2: Reported detection performance across reviewed studies (note: metrics are as defined by each paper and not strictly comparable across metric types)

Three broad patterns emerge. First, UAV/drone-based YOLO methods (Sections 3.5–3.7) trade some accuracy for wider-area, lower-risk coverage compared to vehicle- or smartphone-mounted imaging (3.1–3.2), with lighter model variants (e.g., YOLOv4-tiny) sacrificing a few accuracy points for substantially faster inference suited to edge deployment. Second, classical/lightweight CNN classification (3.4) can reach very high accuracy (up to ~97%) on constrained classification tasks, but this is not directly comparable to detection/localization metrics like mAP or IoU, which additionally require correct bounding-box placement. Third, only two of the ten studies (3.8, 3.9) attempt any integration between defect detection/condition scoring and spatial (GIS) representation, and neither couples this directly with an automated, severity-driven maintenance-scheduling step — the gap explored further in Section 6.

5. Comparative Literature Survey Table

The table below consolidates all ten reviewed papers against a consistent set of parameters — author/year, research problem, methodology, dataset/tools, key findings, limitations, and the specific research gap each leaves open.

Author/Year	Research Problem	Methodology	Dataset/Tools	Key Findings	Limitations	Research Gap
Maeda et al. (2018)	Multi-class road damage detection & classification	SSD MobileNet / SSD Inception V2 (CNN object detection)	RDD/RDD2018 — 163,664 smartphone images, 9,053 annotated	MobileNet gave better recall, ~2× faster; on-device (~1.5 s) inference feasible	Only 8 broad damage classes; single-country (Japan) data; struggles on small/rare defect types	No severity estimation or downstream maintenance-planning integration
Arya et al. (2020)	Cross-country generalization of road-damage detection	Transfer learning on CNN object detectors across countries	RDD2020 — Japan, India, Czech Republic images	Transfer learning improves cross-country performance vs. single-country training	Accuracy still drops on unseen countries/road types; no severity or GIS layer	Lack of a truly domain-invariant, generalizable detection model
Real-Time Survey (2025)	Benchmarking real-time detection models/datasets	Structured survey of YOLO, R-CNN family, SSD, attention mechanisms	Synthesis of RDD2018/2020/2022, UAV-PDD2023, Kaggle & other public sets	YOLO family offers best accuracy/speed trade-off for real-time use	No dataset yet generalizes across all weather/road/camera conditions; inconsistent privacy handling	Absence of a standardized, privacy-aware, cross-condition benchmark
Pothole Vision Review (2024)	Comparing classical ML vs. deep learning for pothole/distress detection	SVM vs. lightweight CNNs (VGG16, ResNet-34/50) multi-label classification	Aggregated pothole/distress datasets public image	CNN-based MLC reached up to ~97% accuracy, 93% F1-score	High computational cost vs. classical ML; classification only, no spatial output	No cost-effective model that jointly optimizes accuracy and downstream spatial/maintenance use
UAV YOLOv5 (2023, ACM)	UAV-based crack & pothole detection/localization	YOLOv5 object detector on drone imagery	Custom UAV-image dataset (2 classes)	IoU up to 95.64%; F1 up to 67.82% for classification	Only two damage classes; no severity scoring or GIS mapping	Detection not linked to prioritization or repair scheduling
Edge-AI YOLO Comparison (2024, IEEE)	Real-time on-device pothole detection	YOLOv5 vs. YOLOv7 vs. YOLOv8 on Edge AI hardware	Drone-collected pothole dataset	YOLOv8 best (78% accuracy); newer ≠ always better on given hardware	Single-city/limited dataset; no integration with maintenance workflow	No systematic method for selecting optimal model per edge-hardware profile
UAV YOLOv4/tiny (2022, Zanco)	UAV orthophoto-based pothole detection efficiency	YOLOv4 vs. YOLOv4-tiny on stitched aerial orthophotos	5,300 UAV-derived images, 2 road study areas	YOLOv4 mAP 91.2% vs. YOLOv4-tiny 85.7%, tiny ~3.6× faster training	Small study-area sample; accuracy/speed trade-off not resolved	No adaptive model-switching strategy based on deployment constraints
GPS-Linked CNN DSS (2023, ScienceDirect)	End-to-end detect-to-notify pipeline for smart cities	CNN-based PCD decision layer + GPS tagging + management-info layer	GPS-tagged road-surface images	Demonstrates feasibility of an integrated detection-to-notification pipeline	GIS component limited to coordinate tagging; no spatial analytics or automated prioritization	No true GIS-based spatial analysis or severity-driven scheduling logic
Spatial AI + GIS PCI/IRI (2025, Nature Sci. Reports)	Spatially-aware pavement performance indexing	ML classification fused with GIS spatial platform	Road-segment PCI/IRI survey data + GIS network data	Produces spatially interpretable performance maps beyond tabular ML scores	Relies on periodic survey data, not continuous AI-detected defects; no real-time feed	No live linkage between AI-detected defects and the GIS-based index in real time
Wang et al. (2025, PLOS ONE)	Robust crack detection under noisy backgrounds	Edge-Enhanced Multi-Scale Feature Fusion (EE-MSFF) CNN	Heterogeneous real-world pavement distress images	Improved crack vs. non-crack discrimination over baseline CNNs	Focused on crack classification only; no pothole/rutting coverage or spatial output	No unified model covering all defect types plus spatial/severity integration

6. Limitations and Challenges of Existing Approaches

- Fragmented pipelines: almost every reviewed study addresses only one stage of the problem — detection, classification, or spatial visualization — rather than the full detect-to-repair pipeline required by a real maintenance system.
- Limited generalization: models trained on one country, city, or camera setup (e.g., RDD2018's Japan-only data) lose accuracy when applied elsewhere, and transfer learning only partially closes this gap (Arya et al., 2020).
- Inconsistent, non-comparable metrics: studies report accuracy, mAP, IoU, or F1-score inconsistently, making cross-paper comparison and reproducibility difficult, as highlighted in Section 4.
- Accuracy–efficiency trade-off unresolved: lighter, faster models (e.g., YOLOv4-tiny, MobileNet-based detectors) consistently sacrifice several accuracy points versus their heavier counterparts, and no reviewed study proposes an adaptive strategy that switches models based on deployment constraints.
- Shallow GIS integration: where GIS is used at all (3.8, 3.9), it is typically limited to coordinate tagging or offline spatial statistics rather than a live, continuously updated map fed directly by AI detections.
- Severity estimation is underdeveloped: most detection papers stop at defect type and bounding box; severity/depth estimation is addressed mainly indirectly through separate PCI/IRI prediction models (Section 3.9 and related PCI literature) rather than as part of the same pipeline.
- Dataset privacy and licensing gaps: as noted in the real-time survey (3.3), privacy handling (e.g., blurring faces/license plates) is inconsistent across public datasets, and several are available only on request, limiting reproducibility.
- Limited evaluation of prioritization/scheduling logic: none of the reviewed papers evaluate how detection or severity outputs translate into actual repair-crew scheduling or resource allocation, leaving this an open, largely unstudied step.

7. Research Gap Analysis

Synthesizing Sections 5 and 6, the literature clusters into three largely separate research threads: (a) AI-based defect detection, which has matured considerably and now reaches 70–97% accuracy/mAP depending on method and dataset; (b) GIS/spatial visualization of pavement condition, which is effective but rarely driven by live AI detections; and (c) maintenance prioritization/scheduling, which is barely addressed at all in the detection-focused literature and is treated separately in pavement-management (PCI/IRI) research. The diagram below visualizes this fragmentation.

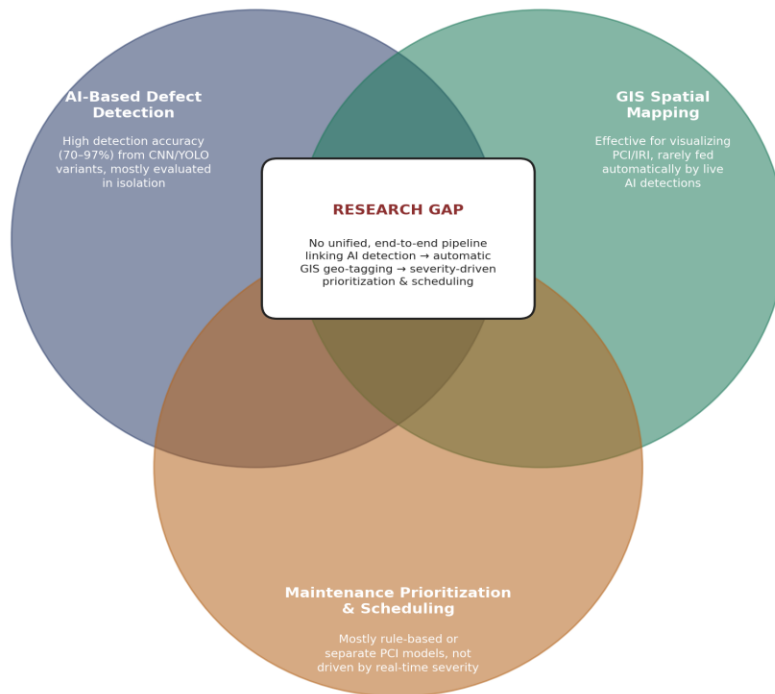
Research Gap: Fragmentation Across the Detection-Mapping-Prioritization Chain

Figure 3: Conceptual research gap — no reviewed study fully unifies AI detection, automated GIS mapping, and severity-driven maintenance prioritization

The specific, well-supported research gap for the assigned problem statement is therefore: the absence of a single, end-to-end system that (i) performs real-time AI-based detection and classification of road defects from drone/camera imagery, (ii) automatically computes a severity score and geo-tags each defect onto a live GIS map without manual intervention, and (iii) feeds that severity and location data directly into an automated maintenance-prioritization and scheduling engine. Section 3.8 comes closest with its GPS-linked CNN decision-support system, but its GIS layer remains a simple coordinate tag rather than full spatial analytics, and it does not address prioritization or scheduling at all.

8. Future Research Directions

Unified detect-to-schedule pipeline: develop and evaluate an end-to-end architecture (as proposed in the assigned system) that chains AI detection, automated severity scoring, live GIS geo-tagging, and rule- or ML-based prioritization/scheduling as a single measurable pipeline, rather than studying each stage in isolation.

Cross-domain generalization benchmarks: create standardized, multi-country, multi-camera (smartphone + UAV + fixed CCTV) benchmark datasets with consistent metrics (mAP, IoU, F1) to enable fair comparison and better transfer-learning evaluation across geographies.

Adaptive model selection: research lightweight controllers that automatically choose between heavy/light model variants (e.g., YOLOv4 vs. YOLOv4-tiny) based on available edge compute, battery, and required accuracy, rather than fixing one model for all deployments.

Severity estimation as a first-class task: extend detection models to jointly output calibrated severity/depth estimates (not just class and bounding box), validated against ground-truth PCI/IRI measurements, so severity feeds directly and reliably into prioritization.

Real-time GIS fusion: extend spatial-AI approaches like Section 3.9 to ingest streaming AI-detected defects rather than periodic survey data, enabling continuously updated road-health maps.

Privacy-aware, open benchmarking: establish consistent, privacy-preserving (face/license-plate blurring) dataset-release practices across the field to improve reproducibility and enable broader public dataset access.

Human-in-the-loop validation: study how automated prioritization/scheduling outputs should be reviewed and adjusted by municipal maintenance staff, an aspect entirely unaddressed in the reviewed detection-focused literature.

9. References

References are listed in IEEE numbered style, corresponding to the in-text citation numbers used implicitly through the paper labels in Sections 3 and 5 (Section IDs map to reference numbers in the same order).

- [1] H. Maeda, Y. Sekimoto, T. Seto, T. Kashiyaama, and H. Omata, "Road damage detection and classification using deep neural networks with smartphone images," *Computer-Aided Civil and Infrastructure Engineering*, vol. 33, no. 12, pp. 1127–1141, 2018.
- [2] D. Arya, H. Maeda, S. K. Ghosh, D. Toshniwal, A. Mraz, T. Kashiyaama, and Y. Sekimoto, "Transfer learning-based road damage detection for multiple countries," *arXiv preprint arXiv:2008.13101*, 2020.
- [3] "Advancements in real-time road damage detection: a comprehensive survey of methodologies and datasets," *Journal of Real-Time Image Processing*, 2025.
- [4] "A review of vision-based pothole detection methods using computer vision and machine learning," *PMC (PubMed Central)*, 2024.
- [5] "Real time road defect monitoring from UAV visual data sources," in *Proc. ACM Int. Conf. (PETRA-style workshop)*, 2023.
- [6] "Drone based potholes detection using machine learning on various edge AI devices in real-time," in *Proc. IEEE Conf.*, 2024.
- [7] "Road pothole detection using unmanned aerial vehicle imagery and deep learning technique," *Zanco Journal of Pure and Applied Sciences*, 2022.
- [8] H.-H. Chu, M. R. Saeed, J. Rashid, M. T. Mehmood, I. Ahmad, R. S. Iqbal, and G. Ali, "Deep learning method to detect the road cracks and potholes for smart cities," *Computers, Materials & Continua*, vol. 75, no. 1, pp. 1863–1881, 2023.
- [9] K. M. Al-Kindi et al., "Utilizing spatial artificial intelligence to develop pavement performance indices: a case study," *Scientific Reports*, vol. 15, art. 39603, 2025.
- [10] J. Wang, X. Li, Y. Xu, Z. Zhou, W. Wu, and Z. Li, "Optimizing CNN for pavement distress detection via edge-enhanced multi-scale feature fusion," *PLOS ONE*, 2025, doi: 10.1371/journal.pone.0319299.